LESSON 5.2

Determine the Area of Triangles

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a triangle using the formula A = ½ × base × height.

WHY THIS MATTERS Your architecture firm is designing a park with triangular garden beds. The client wants to know exactly how much soil to order, which means calculating the area of each triangular section. The first garden has a base of 12 feet and a height of 8 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Base *Spanish: Base*	Any side of a triangle can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.	b in A = ½bh
Height *Spanish: Altura*	The straight-up distance from the base to the top corner.	h in A = ½bh
Area *Spanish: Área*	How much space is inside a flat shape.	measured in cm², in², ft²
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The corner of a book or a door frame — the edges meet at exactly 90°, shown by a small square symbol
Composite figure *Spanish: Figura compuesta*	A shape made by putting two or more simple shapes together.	A house shape = a rectangle (the walls) + a triangle (the roof); total area = rectangle area + triangle area
Formula *Spanish: Fórmula*	A math rule written with symbols.	A = ½ × b × h means Area equals one-half times base times height; for b = 12 and h = 8, A = 48

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A triangle has a base of 14 cm and a height of 9 cm. Find its area.

1. 14 × 9 = 126
2. 126 is the parallelogram made by two triangles.
3. 126 ÷ 2 = 63

Answer: 63 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller triangle: base = 10 cm, height = 6 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base of the triangle is _____ and the height is _____ .
- I know that measure is the height because _____ .
- Two copies of this triangle would compose a _____ .
- Area = ½ × _____ × _____ = _____ square units.

Word bank triangle | parallelogram | half | base | height | perpendicular | square corner | area | formula | right triangle | copy | rotate

WATCH OUT!

A common mistake

A common mistake in Area of Triangles is forgetting the ½

LESSON 5.2

Determine the Area of Triangles

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use the formula A = ½ × b × h to find the area of each triangle.

Triangle	Base	Height	Area
Sail A	6 ft	8 ft	
Sail B	10 ft	5 ft	

HINT Each sail row gives one base and one height — the blank Area column is yours to fill.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 10 cm and height 6 cm?

(A) 16 sq cm
(B) 30 sq cm
(C) 30 cm
(D) 60 sq cm

HINT You have a base of 10 cm and a height of 6 cm — a triangle needs only those two, plus one special step.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A triangular garden bed in the firm's park design has an area of 24 sq ft and a base of 8 ft. What is the height?

(A) 3 ft
(B) 6 ft
(C) 12 ft
(D) 16 ft

HINT The area (24 sq ft) and base (8 ft) are given — you're solving backwards for the height.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

Why do we divide by 2 when finding the area of a triangle?

(A) A triangle is exactly half of a rectangle with the same base and height

(B) Triangles have 2 equal sides

(C) The base is always twice the height

(D) We always divide area formulas by 2

HINT Think about drawing a diagonal across a rectangle — what two shapes does it create?

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the area of a triangle with base 14 ft and height 6 ft?

(A) 20 sq ft

(B) 42 sq ft

(C) 42 ft

(D) 84 sq ft

HINT Base 14 ft and height 6 ft — remember what makes a triangle's area different from a rectangle's.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Identify base and height** base = 9 ft, height = 4 ft
2. **Write the formula** A = ½ × base × height
3. **Multiply base × height** 9 × 4 = 36
4. **Final answer** A = 36 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at the formula: A = ½ × base × height. Does the final answer include the ½?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

SENTENCE STARTER The area of the triangle is ___ the rectangle because A = 1/2 × ___ × ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.2

Determine the Area of Triangles

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Triangle	Base	Height	Area
Garden A	12 ft	8 ft	
Garden B	9 ft		27 sq ft
Garden C		10 ft	35 sq ft
Garden D	15 ft	4 ft	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each shape to its area. Watch the rule: triangles use A = ½ × base × height, but parallelograms and rectangles use A = base × height (do NOT take half).

_____	Triangle b=8, h=6	(A)	24 sq units
_____	Triangle b=10, h=8	(B)	40 sq units
_____	Triangle b=12, h=5	(C)	30 sq units
_____	Parallelogram b=5, h=4	(D)	20 sq units
_____	Rectangle 7×3	(E)	21 sq units
_____	Parallelogram b=9, h=2	(F)	18 sq units

3 **Write the name of the correct group on each line.**

Calculate the area of each triangle. Sort by whether the area is greater than 30 sq units or not.

Groups: **Area > 30 sq units** **Area ≤ 30 sq units**

_____ b=10, h=8

_____ b=6, h=9

_____ b=14, h=5

_____ b=8, h=7

_____ b=12, h=6

_____ b=10, h=6

4 **Mark and label each value on the number line.**

Find the missing height: b = 10, A = 35



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Identify base and height** base = 12 ft, height = 7 ft
2. **Write the formula** A = ½ × base × height
3. **Substitute values** A = ½ × 12 × 7
4. **Calculate** A = (½ × 12) × (½ × 7) = 6 × 3.5 = 21 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.2

Determine the Area of Triangles

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Use the model to solve. Show your work.**



Rectangle 16×10 = 160 sq ft split into two equal triangles of 80 sq ft each

SHOW YOUR WORK

Answer: _____

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Area Error

1. **Identify base and height** b = 14 m, h = 9 m
2. **Write the formula** A = ½ × b × h
3. **Substitute values** A = ½ × 14 × 9
4. **Calculate** A = ½ × 23 = 11.5 sq m

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A rectangle is 12 ft by 8 ft. A diagonal line cuts it into two triangles. What is the area of each triangle? Explain how the triangle area formula relates to the rectangle area formula.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help